

Grant application pre-proposal form 2024

NWO Talent Programme – Veni scheme

Applied and Engineering Sciences
Science
Social Sciences and Humanities



Version: June 2024

1. Institution and field of research

1a. NWO domain (Choose one)

Applied and Engineering Sciences (AES/TTW)

1b. Main field of research

Indicate the main field of research and (if applicable) other fields of research, in order of relevance, using the codes and names from the dropdown menu. For more information see the Notes.

	Code/Field of research:
Main field of research:	16.90.00 Computer science, other
Other field(s) of research (if applicable):	Choose second field of research (if applicable)
	Choose third field of research (if applicable)
	Choose fourth field of research (if applicable)

2. Evidence Based Curriculum Vitae

2a. Academic profile

Cybersecurity is paramount in today's societies, with services that heavily rely on the Internet to function. For example, the NWO Isaac system that got attacked in the week of August 19, 2024. When starting my PhD in cybersecurity in 2018 I was not planning on staying in academia, but rather to help combat these attacks in the real world. Indeed, after my PhD I left for the Ministry of Defence to tackle real-world cybersecurity problems. But after 1,5 years I realized that some problems society faces require an academic approach. This inspired me to return to TU Delft in 2024, to help bridge the gap in both industry and academia using my experiences on both sides.

My research is on Cyber Threat Intelligence (CTI): I try to understand the behaviour of attackers, learn their techniques, their objectives and use that intelligence to deploy defences in the most useful places as all attackers have different methods and goals (fig.1). I use large networks of "fake devices", so-called honeypots, to gather data from real attacks that happen on the internet – enabling me to analyse their patterns. This has proved successful. I investigated the behaviour of the largest botnet to date, *Mirai*, looking at 300M attacks[1] and tactics in DDoS attacks[2]. During these studies, **I learned that while most attackers rely on simple techniques, a small number of attackers show a deep understanding of attack techniques. While measuring cyber threats, research is heavily biased to the large number of "simple" attackers, but I believe that we should instead focus on advanced threats.** I was invited to present this work at the Dutch nuclear security operators meeting – a sector where cybersecurity is of utmost importance - and at *Intersct2023*[3], a conference attended by practitioners and academics. During discussions resulting from these talks, **I have initiated collaborations with nuclear facilities in the Netherlands** to help these organizations get a better view of their threat landscape, while simultaneously allowing me to investigate attacks and better understand the abilities of advanced adversaries. **I have also initiated active collaborations with the ministry of Defense.**

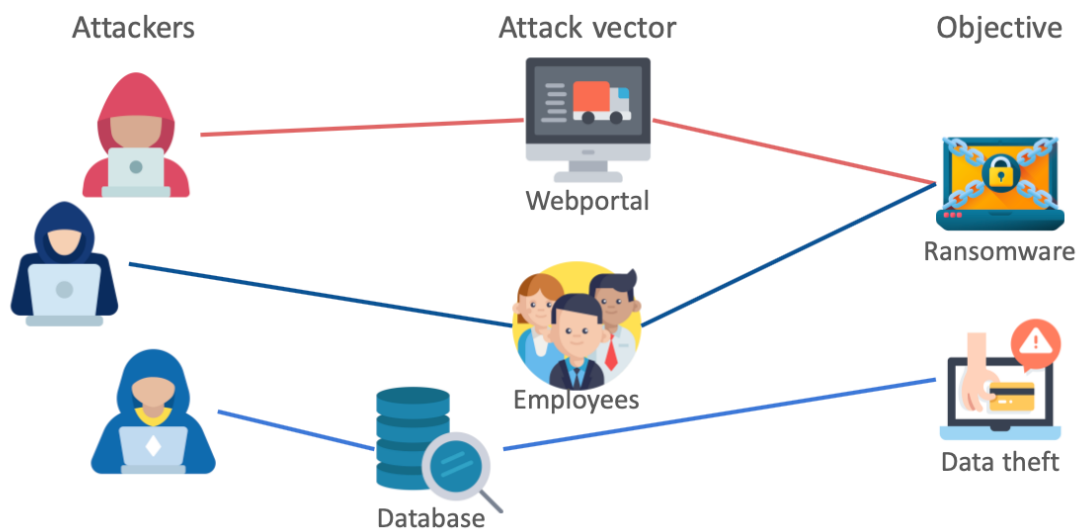


Figure 1: Attackers use different methods and have different goals

The analysis of attacker behaviour also led to industrial interest for my results. I have evaluated the effectiveness of state-of-the-art practices in CTI in use by large organisations such as the Ministry of Defence, Dutch banks and companies like Fujitsu[5,6]. Following this work, **I was invited to visit Fujitsu to collaborate on a project where we managed to temporarily takeover a newly discovered botnet[10].** In addition, I was approached by Prof. Feldmann (Director, Max Planck Institut für Informatik) to develop a method to capture unsolicited Internet traffic[7], which is **being integrated in the largest Internet Exchange Provider in Europe.**

Being able to connect an academic mindset and an industrial focus allows me to come up with original solutions, as evidenced by output [8] which shows that by monitoring malicious internet traffic it is possible to gain better insight into network stability. I created a tool based on this method, *churn-monitor*[9], which allows cyber threat analysts to better understand the internet and make more informed decisions. My work is also applicable on a larger scale. **I developed a novel way of dealing with botnets**, reducing the rate in which they spread by 50%, thereby improving the safety of the Internet[4].

My position as an expert in the field is underlined by the invitation to join the Program Committee for USENIX Security, the largest academic security conference.

Now back in academia, I'm excited to build my group on Cyber Threat Analysis and Cyber Deception. Over the course of my academic career, I have supervised 12 Msc students and have recently recruited three PhD students, who are in their first six months of working. I also enjoy teaching, and during my PhD I was a part-time lecturer. I am currently the responsible lecturer for Computer Security, a course for Bachelor students, and I am adding CTI in the master programme. With the Veni I am to take a novel step in Cybersecurity: **move from merely analysing patterns in cyberattacks to proactively predicting future behaviour and deploy defenses.**

Word count 2a Academic Profile: 700

2b. Key output

Key output 1

Open Access: Yes

Reference: **Griffioen, Harm**; Doerr, Christian. Examining Mirai's battle over the internet of things, Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security, 743-756, 2020

URL: <https://doi.org/10.1145/3372297.3417277>

Type: Conference paper

Quality indicators (only add additional information in the motivation section below):

- 1) Citations: Article Field Weighted Citation Impact
- 2) Academic invitations directly related to this output
- 3) Optional: choose a third indicator

Motivation: I investigated the largest botnet to date using a set of novel datasets consisting of more than 9.000 sensors. This resulted in a conference paper, which is the most important output in my domain of cybersecurity. This paper has high relevance and impact in the field of cybersecurity, as evidenced by its significant **FCWI of 6.06**. Due to this publication, I have been **invited to provide a talk at the Nuclear security convention** attended by the heads of security of all Dutch nuclear facilities.

Key output 2

Open Access: Yes

Reference: **Griffioen, Harm**; Oosthoek, Kris; van der Knaap, Paul; Doerr, Christian. Scan, test, execute: Adversarial tactics in amplification DDoS attacks, Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security, 940-954, 2021

URL: <https://doi.org/10.1145/3460120.3484747>

Type: Conference paper

Quality indicators (only add additional information in the motivation section below):

- 1) Academic invitations directly related to this output
- 2) Targeted impact, has reached a specific audience
- 3) Citations: Article Field Weighted Citation Impact

Motivation: I investigate adversarial tactics in one of the largest threats facing the Internet to better understand how we should build our defenses. This resulted in a conference publication that has a **FWCI of 2.71**, and led to an **invitation to talk at the Intersct conference**. The dataset generated during this experiment has **more than 100x the amount of measurement points of previous studies** and has been requested by 5 research groups.

Key output 3

Open Access: Yes

Reference: Intersct 2023 invited talk: Scan, test, execute: Adversarial tactics in amplification DDoS attacks

URL: <https://intersct.nl/event/intersct23/>

Type: Invited talk

Quality indicators (only add additional information in the motivation section below):

- 1) Academic interest, other
- 2) Industry use/interest
- 3) Optional: choose a third indicator

Motivation: I was happy to receive an invitation by *Prof. dr. Herbert Bos, Hoogleraar VU Amsterdam* to present my work at Intersct 2023, an event aimed at both the academic and industrial communities in cybersecurity. After the talk, several *Dutch ministries* inquired about my research and the implication of my findings. The interest of practitioners inquiring about the research highlights the practical applications and impact of the presented research, demonstrating a **bridge between theoretical research and real-world implementation**.

Key output 4

Open Access: Yes

Reference: **Griffioen, Harm**; Doerr, Christian. Could you clean up the Internet with a Pit of Tar? Investigating tarpit feasibility on Internet worms, 2023 IEEE Symposium on Security and Privacy (SP), 2551-2565, 2023, IEEE

URL: <http://doi.org/10.1109/SP46215.2023.10179467>

Type: Conference paper

Quality indicators (only add additional information in the motivation section below):

- 1) Civil-society effects
- 2) Originality/novelty
- 3) Optional: choose a third indicator

Motivation: During this study, I single handedly slowed down the largest botnet to date by 50%. This means I prove that I can successfully decrease the effectiveness of this botnet, which is responsible for devastating DDoS attacks on Twitter, Reddit, Netflix, and Airbnb. With my approach I demonstrate a **different approach to cyber defense strategies**.

Key output 5

Open Access: No

Reference: **Griffioen, Harm**; Booij, Tim; Doerr, Christian. Quality evaluation of cyber threat intelligence feeds, Applied Cryptography and Network Security: 18th International Conference, ACNS 2020, Rome, Italy, October 19–22, 2020, Proceedings, Part II 18, 277-296, 2020, Springer International Publishing

URL: https://doi.org/10.1007/978-3-030-57878-7_14

Type: **Conference paper**

Quality indicators (only add additional information in the motivation section below):

- 1) Academic invitations directly related to this output
- 2) Citations: Article Field Weighted Citation Impact
- 3) Industry use/interest

Motivation: In this work I showed the quality – and with that the effectiveness – of Cyber Threat Intelligence feeds used by organizations. The work was well received in the community (**FCWI 4.03**) and its results have been applied in industrial settings (*Fujitsu, several Dutch ministries*). In addition, I was invited to collaborate with Prof. dr. Michel van Eeten to do a follow up study.

Key output 6

Open Access: Yes

Reference: Bouwman, Xander; **Griffioen, Harm**; Egbers, Jelle; Doerr, Christian; Klievink, Bram; Van Eeten, Michel. A different cup of TI? The added value of commercial threat intelligence, 29th USENIX security symposium (USENIX security 20), 433-450, 2020

URL: <https://www.usenix.org/system/files/sec20-bouwman.pdf>

Type: **Conference paper**

Quality indicators (only add additional information in the motivation section below):

- 1) Academic collaboration and/or interdisciplinary engagement
- 2) Industry use/interest
- 3) Citations: Article Field Weighted Citation Impact

Motivation: This work resulted from an **invitation to collaborate** from *Prof. dr. van Eeten, hoogleraar TU Delft*. The paper explores the unique contributions of commercial threat intelligence and investigates whether the quality of the information meets the expectation of analysts. Organizations like *Cato networks* have shown interest in applying the findings to enhance their cyber threat intelligence capabilities, and the work was mentioned on the *prnewswire*. I was responsible for the visualization, formal analysis, validation, and review & editing. This paper has an **FWCI of 7.53**.

Key output 7

Open Access: Yes

Reference: Wagner, Daniel; Ranadive, Sahil Ashish; **Griffioen, Harm**; Kallitsis, Michalis; Dainotti, Alberto; Smaragdakis, Georgios; Feldmann, Anja. How to Operate a Meta-Telescope in your Spare Time, Proceedings of the 2023 ACM on Internet Measurement Conference, 328-343, 2023

URL: <https://doi.org/10.1145/3618257.3624831>

Type: **Conference paper**

Quality indicators (only add additional information in the motivation section below):

- 1) Academic collaboration and/or interdisciplinary engagement
- 2) Industry use/interest
- 3) Optional: choose a third indicator

Motivation: I was invited to collaborate on this paper led by *Prof. Feldmann, Direktor, Max Planck Institut für Informatik*. due to my works in Internet measurements. This paper demonstrates how Internet traffic can be used to create network vantage points in different locations where you are not able to establish a normal vantage point. **The findings have been integrated into industrial products** by the largest Internet exchange point in Europe. During this project, I was responsible for a part of the data curation, formal analysis, methodology, and review & editing.

Key output 8

Open Access: No

Reference: **Griffioen, Harm**; Doerr, Christian. Quantifying autonomous system IP churn using attack traffic of botnets, Proceedings of the 15th International Conference on Availability, Reliability and Security, 1-10, 2020,

URL: <https://doi.org/10.1145/3407023.3407051>

Type: **Conference paper**

Quality indicators (only add additional information in the motivation section below):

- 1) Reproducibility
- 2) Originality/novelty
- 3) Optional: choose a third indicator

Motivation: I devised a novel method to use malicious network traffic to gain insight in network stability leveraging botnet traffic for benign purposes. The work has shown to be reproducible by other researchers that have

implemented the method, eliminating a long-standing problem in Internet measurements where it was not possible to account for changes in a network.

Key output 9

Open Access: Yes

Reference: Churn monitor
URL: https://harmgriffioen.github.io/churn-monitor/
Type: Software
Quality indicators (only add additional information in the motivation section below): 1) Targeted impact, has reached a specific audience 2) Industry use/interest 3) Optional: choose a third indicator

Motivation: The Churn Monitor software addresses a specific need in network management: monitoring IP churn. It has gained notable industry interest, particularly from cybersecurity analysts, who use it to improve their understanding of the network landscape and the reliability of their findings.

Key output 10

Open Access: No

Reference: Taniguchi, Tsuyoshi; Griffioen, Harm ; Doerr, Christian. Analysis and takeover of the bitcoin-coordinated pony malware, Proceedings of the 2021 ACM Asia Conference on Computer and Communications Security, 916-930, 2021
URL: https://doi.org/10.1145/3433210.3437520
Type: Conference paper
Quality indicators (only add additional information in the motivation section below): 1) Industry use/interest 2) Academic collaboration and/or interdisciplinary engagement 3) Optional: choose a third indicator

Motivation: In this work I helped take over a live botnet that was using a method to exchange commands that was never seen before, temporarily eradicating the impact of the network as attackers were unable to perform their malicious activities. This work resulted from an **invitation to visit Fujitsu laboratories and collaborate with their researchers**. During this work I was responsible for the visualization, formal analysis, validation, and review & editing, as well as the actual takeover of the botnet.

Word count 2b Key output: 700

3. Research idea

Title: Advancing Cyber Threat Intelligence: Leveraging Honeypots and Telemetry for Proactive Defense Strategies
Key words (max 5): Proactive Cybersecurity, Threat Intelligence, Deception

Research idea:

Cyberattacks are increasingly prevalent and sophisticated, outpacing traditional reactive cybersecurity strategies. Sophisticated malware (e.g. COATHANGER) is often discovered when damage is already done. To defend ourselves, we must adopt proactive cyber defenses, where we understand how attackers operate, their goals, and how their behavior evolves. My research leverages deception techniques by deploying vulnerable systems to attract real-world attacks. I will analyze the collected data to identify patterns, develop predictive models to anticipate future threats, and refine defense mechanisms based on evolving attacker strategies. This approach aims to enhance cybersecurity resilience by staying ahead of adversaries.

Word count 3c Research idea: 95

4. Administrative details

4a. Personal details

Title(s), initial(s), surname(s):	Dr. Ir. H.J. Griffioen
E-mail address:	h.j.griffioen@tudelft.nl
Phone number:	0628213178
Mobile phone number:	0628213178

4b. Master's degree ('doctoraal')

University/College of Higher Education:	TU Delft
Main subject:	Computer Science

4c. Doctorate

University/College of Higher Education:	TU Delft
Starting date (dd/mm/yy):	01/11/2018
Date of PhD award (dd/mm/yy):	30/09/2022
Supervisor(s) ('Promotor(en)'):	Christian Doerr, Inald Lagendijk
Thesis title:	Cyber Threat Intelligence: Analysis of Adversaries and their methods

4d. Prospective host institution

Host institution:	TU Delft
Research group:	Cyber Security (CYS)

4e. Work experience since completing your (first) PhD

List your appointments chronologically. The bottom row should contain your current position.

Position	Period (date-date)	FTE	Position type (fixed term/permanent/tenure-track/other)	Institution
Cyber Threat Analyst	01/06/2022 – 31/12/2023	1	Permanent	Ministerie van Defensie
Postdoc	01/10/2022 – 31/12/2023	0.2	Fixed term	TU Delft
Assistant Professor	01/01/2024 – 31/05/2024	1	Tenure-track	TU Delft
Assistant Professor	01/06/2024	1	Permanent	TU Delft

4f. Net. academic research time

Calculate your net academic research time, as described in the notes. Do not include the calculation, just give the actual number of months.

Number of months:	13
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If applicable: You may mention special circumstances that account for a reduction in productivity (max. 100 words):

Statements by the applicant

Use of extension clause

If you make use of the extension clause, (only) add the date of the confirmation e-mail (talent@nwo.nl, previously: vi@nwo.nl) that the extension was granted. An extension is only necessary if you exceed the year limit on the reference date.

Do you make use of the extension clause: No

If yes, the extension was confirmed on: Choose the date

By submitting this form I declare that:

By submitting this form, I declare that I have completed this form truthfully, and that I satisfy the nationally and internationally accepted standards for scientific conduct as stated in the Netherlands code of conduct for research integrity 2018. I declare that my contact details in ISAAC profile are up to date (i.e. postal address, phone number, personal details and correspondence language). I endorse the code of conduct for laboratory animals and the code of conduct for biosecurity/possibility for dual use of the expected results and will act accordingly, if applicable.

Initial(s) and surname(s)¹: H.J. Griffioen

Place: Delft

Date: 31-08-2024

¹ Please refrain from using your first name to reduce gender effects.